Explainable Fake News Detection

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Machine Learning (CISC-5800)
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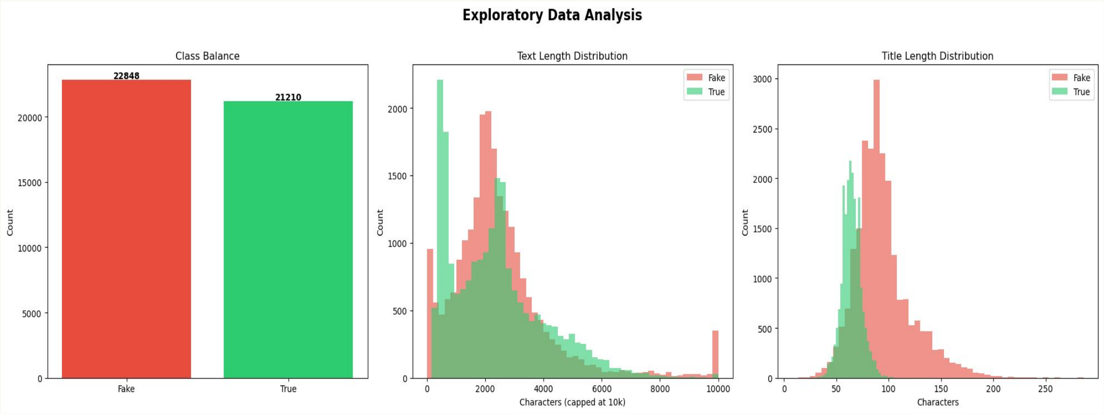
Problem Statement and Dataset

- Fake news spreads rapidly online, causing real-world harm
- Goal: Build a transparent classifier that detects fake news and explains its decisions
- Dataset: 44,898 news articles (ISOT Research Lab UVi)
    - fake.csv (23k) - Politifact & Wikipidea flagged
    - true.csv (21k) - News Websites crawling
- Binary classification: REAL (1) vs. FAKE (0)
- Columns: title, text, subject, date, **label**

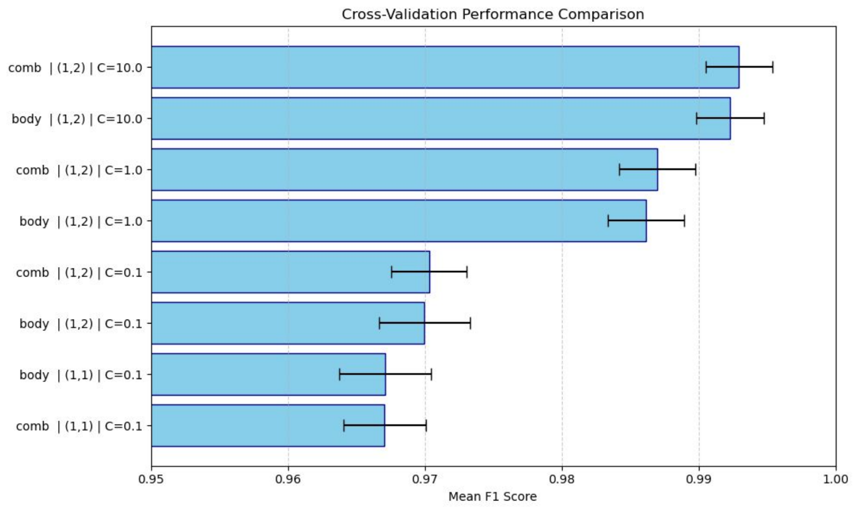
Data Preprocessing

- Missing value samples and Duplicates dropped
- Raw text contained noise: URLS, punctuation, stopwords,
- Cleaning pipeline:
  - Lowercasing
  - Removed URLs, twitter handles (bias), special characters
  - Removed stopwords (NLTK)
  - Lemmatization (WordNetLemmatizer)
- Stored cleaned text in cleaned_text column
- Split: 75% train / 25% test, stratified to preserve class balance



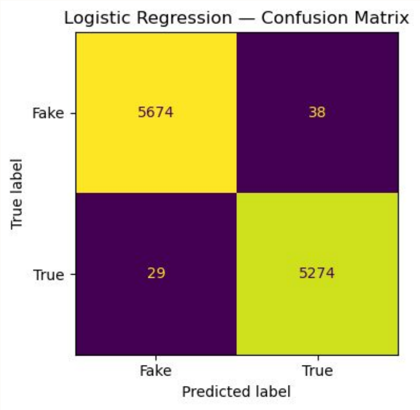
Feature Selection

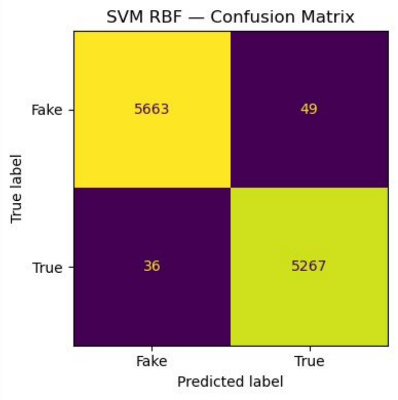
- Compared two feature representations:
  - **Term frequency-Inverse Document Frequency (TF-IDF)**
    i. 50k L2 normalized sparse features representing article bigrams
    ii. Used 5 fold CV for hyperparameter tuning **(C, ngram_range)** and feature selection **(body or title+body)**
    iii. F1: **0.9951**
  - **Word2Vec** (avg. pooling, 100 dims)
    i. Word => 100-dimensional vector
    ii. Collapsed to 100-dimension vector per sample
    iii. F1: 0.9749
- TF-IDF won on unseen data
- Best configurations: bigrams (1,2), 50k features, titile+text

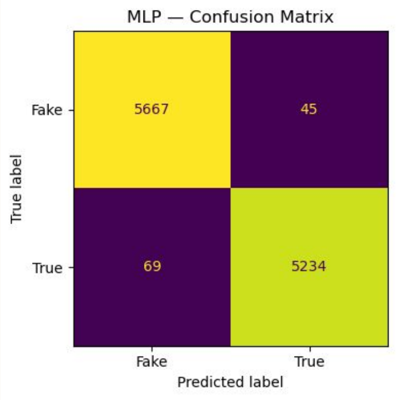


Model Training and Performance

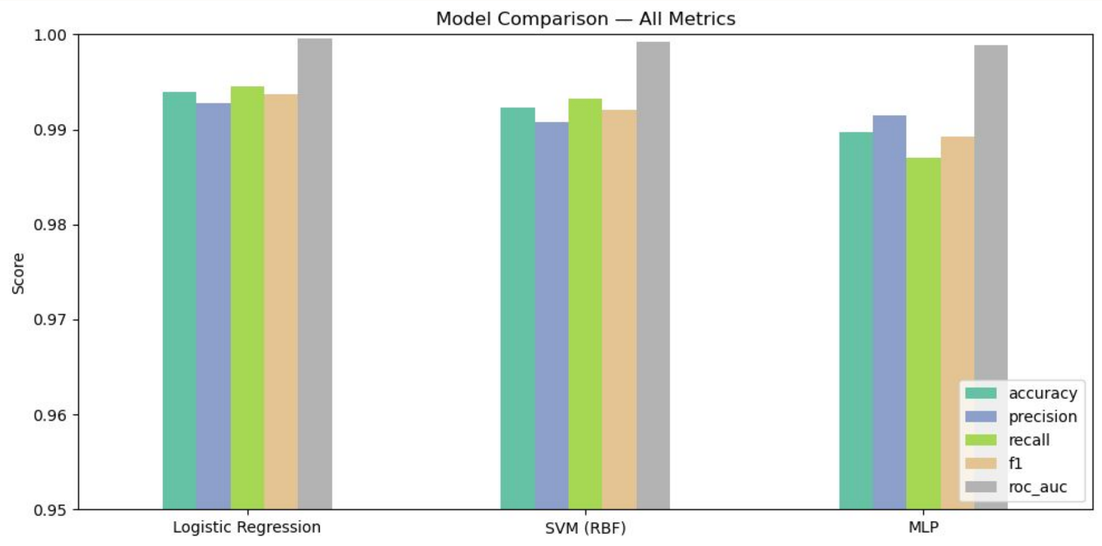
- Three models trained and compared:
  - **Logistic Regression** (used full TF-IDF 50k dims)
  - **Support Vector Machine** (used TruncatedSVD 300 dims)
    - RBF doesn't scale to 50k sparse dims
    - Slowest
  - **PyTorch MLP** (used TruncatedSVD 300 dims)
    - 3-layer, Sigmoid activation, Adam Optimizer, BCE loss, lr=1e-3, dropout=0.3
- Best model: Logistic Regression - fast, interpretable, highest metrics





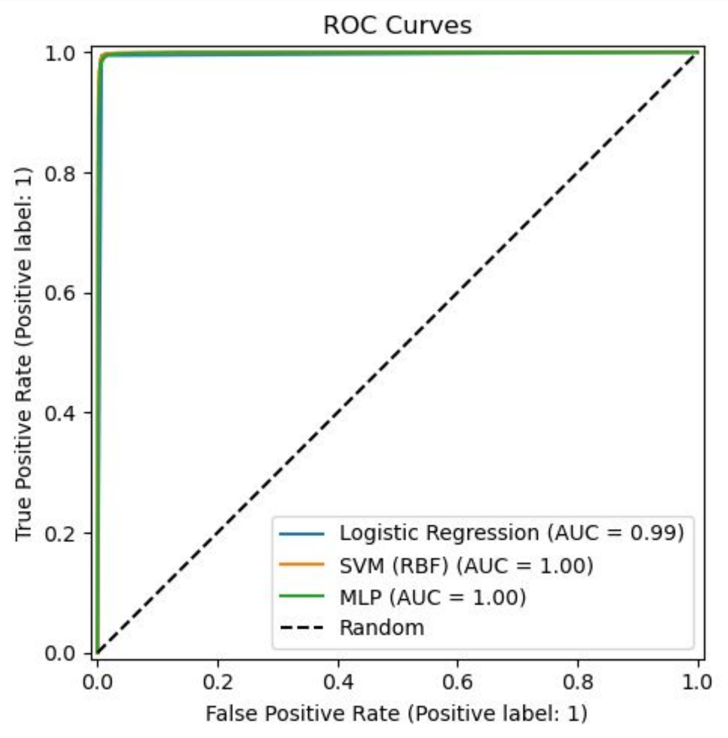


Model Training and Performance



=== Model Comparison ===

	accuracy	precision	recall	f1	roc_auc
Logistic Regression	0.9939	0.9928	0.9945	0.9937	0.9996
SVM (RBF)	0.9923	0.9908	0.9932	0.9920	0.9992
MLP	0.9897	0.9915	0.9870	0.9892	0.9989



RAG Explainability

- Problem: classifiers are black boxes giving a label, not a reason
- Solution: Retrieval-Augmented Generation (RAG) pipeline
- Pipeline:
  - Retrieve - **Facebook AI Similarity Search (FAISS)**
    - Optimized vector retrieval, opensource
    - finds the k most similar training articles (cosine similarity)
  - Extract - top TF-IDF phrases identify the most influential words **our model values**
  - Explain - template prompt **+ google/flan-t5-large** LLM generates a natural language explanation
    - Pretrained model downloaded from HuggingFace and ran locally
- Example output on sample test article:

  “””
  True: FAKE | Predicted: FAKE ✓

  Explanation: *This article is classified as FAKE with 98.8% confidence. The most distinctive phrases found were: 'trump', 'blocked', 'twitter', 'ben jerry', 'flavor'. These phrases appear frequently in FAKE articles in the training data. 2 out of 3 most similar training articles were also labeled FAKE, further supporting this prediction. Topic summary: trump blocks twitter*
  “””

References

- ISOT Dataset - https://onlineacademiccommunity.uvic.ca/isot/2022/11/27/fake-news-detection-datasets/
- Project Implementation - https://github.com/natitedros/transparent-fake-news-detector

Thank you!